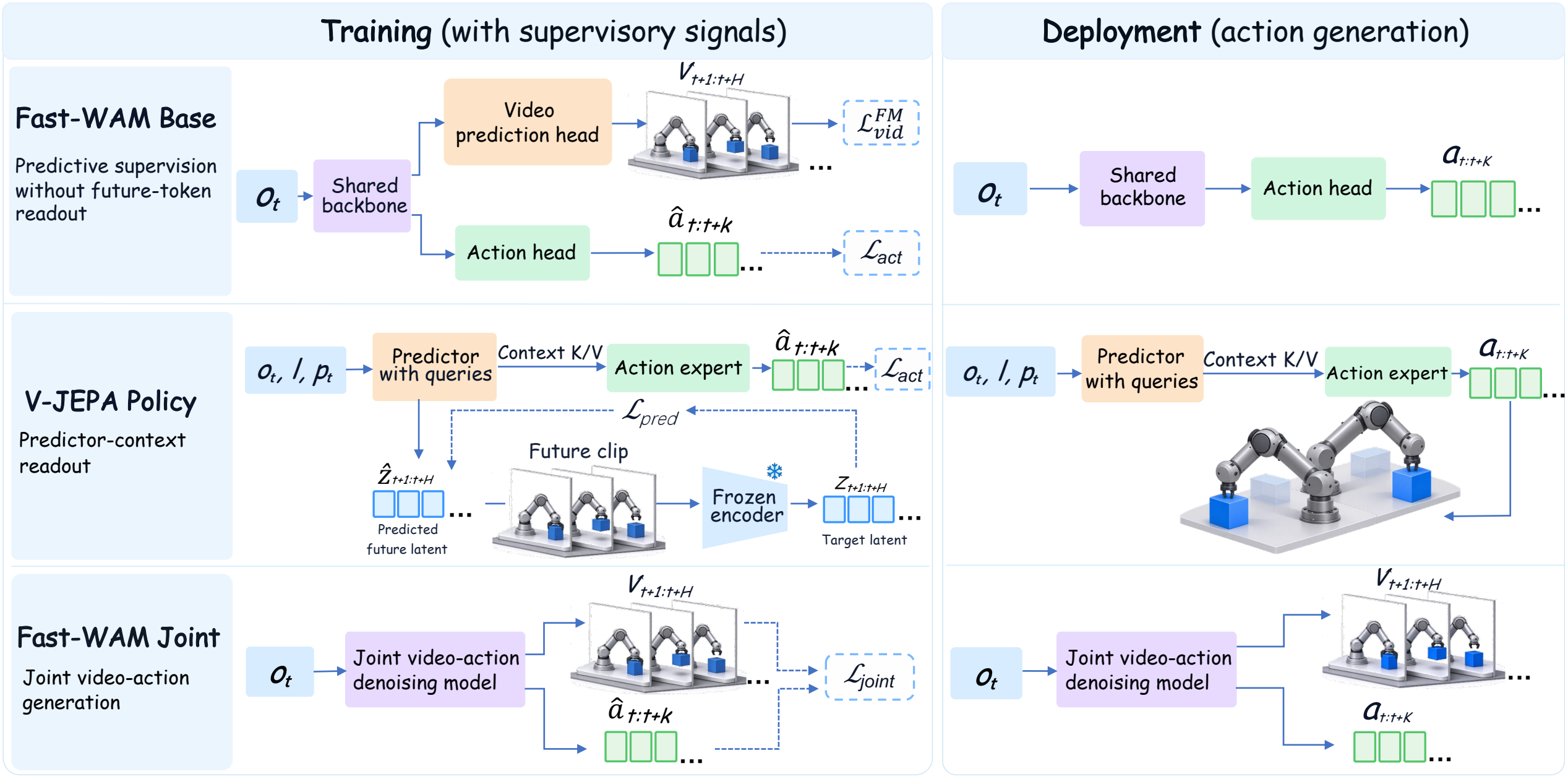

Training (with supervisory signals)
Deployment (action generation)
Fast-WAM Base
Predictive supervision without future-token readout
o_t
Shared backbone
Video prediction head
$V_{t+1:t+H}$
$\mathcal{L}_{vid}^{FM}$
Action head
$\hat{a}_{t:t+k}$
$\mathcal{L}_{act}$
o_t
Shared backbone
Action head
$a_{t:t+K}$
V-JEPA Policy
Predictor-context readout
o_t, l, p_t
Predictor with queries
Context K/V
Action expert
$\hat{a}_{t:t+k}$
$\mathcal{L}_{act}$
$\mathcal{L}_{pred}$
Future clip
$\hat{z}_{t+1:t+H}$
Predicted future latent
Frozen encoder
$z_{t+1:t+H}$
Target latent
o_t, l, p_t
Predictor with queries
Context K/V
Action expert
$a_{t:t+K}$
Fast-WAM Joint
Joint video-action generation
o_t
Joint video-action denoising model
$V_{t+1:t+H}$
$\hat{a}_{t:t+k}$
$\mathcal{L}_{joint}$
o_t
Joint video-action denoising model
$V_{t+1:t+H}$
$a_{t:t+K}$
Forward computation
Supervisory signal (training only)
Observation/context
Prediction-related module
Action output
Shared/reused module (not necessarily frozen)